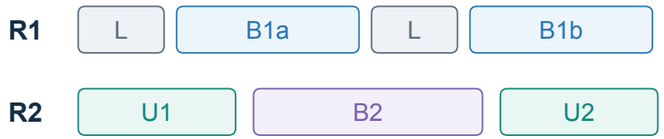
TagForge

From complex barcode sequencing to molecular quantification

Parse structure → correct barcodes → count molecules

01 Parse library structure

B1: identity (e.g. cell / sample); B2: feature



Linker localization; fixed fallback if configured

02 Correct each segment

Per-segment whitelist correction

Mismatch A C G A → A C G T

Shift T A C G T G → A C G T

Correct segments, then assemble barcodes

03 UMI clustering & counting

Group by identity barcode × feature

12 reads → **2** molecules

04 Molecule evidence → sparse counts

B1	Feature	UMI	Reads
B01	F1	AAAC	9
B01	F1	GGGC	3

→ **2** molecules (B01 × F1)

Parquet · MTX · TSV

05 Downsampling diagnostics

Downsample molecule-supporting reads

Sequencing saturation
Read redundancy

Non-singleton fraction
Fraction with ≥2 reads

APPLICATIONS & EXTENSIONS

Identity barcode × feature barcode × UMI

CONFIGURE · EXTEND

- **Antibody & combinatorial tags** — Identity barcode × antibody tag
- **Sample Tags** — Cell × sample tag
- **CRISPR Guide Capture** — Cell × sgRNA
- **Antigen Capture** — Cell × antigen tag
- **Promoter / reporter barcodes** — Identity barcode × design tag

L: linker · U: UMI. Illustrative layout and counts; extensions require validation.

TagForge

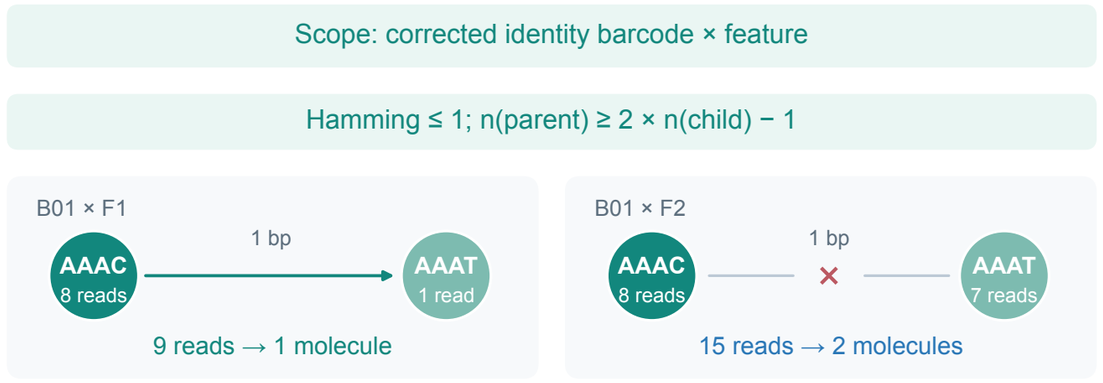
Core mechanisms: barcode identity and UMI-defined molecules

Whitelist correction resolves identity; UMI clustering collapses reads.

A Barcode identity correction



B UMI molecule grouping



ENGINEERING ADVANTAGES

Scale and optimize computation

Block reads · Parquet · DuckDB · sparse output

Exact matching · neighbor lookup · correction cache

Parallelism across independent groups

Bucket complete barcode–feature pairs; preserve UMI clustering boundaries.

Recover and validate results

Resume committed extraction / correction batches.

Validate inputs, config and upstream state before reuse.

Batch execution and progress tracking

Slurm · live progress · segment QC

Sample and batch HTML / Excel reports

UMI-tools directional, default distance 1; illustrative counts.

TagForge

Two dedicated libraries: molecular identity and cell linkage

PB: shared identity barcode; FB: antibody feature barcode; CB: cell barcode.

Shared core: structure parsing → barcode handling → UMI clustering → supporting reads

A PI-seq: assign molecules to PBs

Which PB carries this antibody-tagged molecule?

R1 PB1 | L | PB2 | L | PB3 | L | PB4 | L | PB5

R2 U1 (8 nt) | FB (12 nt) | U2 (8 nt)

Within each FB × corrected UMI: compare PB read support

FB1 × UMI-a

PB01 : PB02 = 9 : 1

90% → retain PB01

FB2 × UMI-b

PB01 : PB02 = 8 : 2

80% → exclude group

Single-PB groups are retained directly.

Output: PI-filtered PB × FB molecule details

molecule_detail.rmMP.parquet

B PB–CB: map PBs to cells

Which cell barcode is supported by this PB?

PB × Observed CB × UMI

No CB whitelist; optional conservative observed-CB correction.

For each PB: aggregate CB read support after CB handling

PB01

CB01 : CB02 = 9 : 1

90% → map to CB01

PB02

CB01 : CB02 = 8 : 2

80% → no assignment

Read support selects the CB; UMI counts remain available.

Output: PB → CB map and PB counts per cell

pb_cb_map.tsv.gz · cb_pb_counts.tsv.gz

LINK THE TWO RESULTS THROUGH SHARED PBs

PI-seq: PB × antibody feature + PB–CB: PB → cell → Downstream: cell-associated feature evidence

The two outputs can be joined on PB; downstream integration is performed separately.

Examples only. Dominant reads / total reads > threshold (default 0.8); layout is configurable.